

## **TAPER**

Definition: An object whose diameter is gradually increased or decreased at a uniform rate along a given line or plane is called taper.

### **Necessity of taper :-**

1. Taper ensures self alignment of the tool like drill, reamer to the axis of the machine spindle.
2. For quick fixing
3. For quick releasing
4. For easy assembly and dismantle of parts.
5. For transmitting the drive in the assembled parts.

### **Expression of taper :-**

1. By giving Big dia (D), small dia (d) & taper length (l) 180 mm.  
e.g.  $D = 40 \text{ mm}$   $d = 30 \text{ mm}$  and  $l = 180 \text{ mm}$
2. By taper per foot, e.g.  $5/8 \text{ inch T.P.F}$
3. By Taper per inch, e.g.  $0.1 \text{ inch T.P.I}$
4. By degree, e.g.  $60^\circ$ .
5. By ratio, e.g. 1: 20 (1 unit taper (D-d) on 50 unit length)
6. By conicity, e.g.  $1/200$ .
7. By standard taper, e.g. MT-3, M-80 taper etc.

## **TAPER TURNING**

Taper turning :- When the turning operation is performed at an angle with the job axis, is called Taper turning.

### **Principle in setting :-**

1. Setting the job aligning with the lathe axis and the cutting tool moves at an angle.
2. Setting the job at angle and the cutting tool travels parallel to the axis of the lathe.

### **Methods of taper turning :-**

1. By offsetting the tail stock.
2. By swivelling the compound rest.
3. By taper turning attachment.
4. By formed tool.

### **1.TAPER TURNING BY OFF SETTING TAIL STOCK(SET-OVER METHOD)**

Principle in setting:- The job is not at an angle and the cutting tool moves parallel to the lathe axis.

#### **Method of off setting:-**

- In this method tail stock is set out of alignment.



- Off setting is done by shifting the body of the tail stock over the base by means of two screws, provided on the tail stock body.
- The body may be slid in both the directions, i.e., towards the operator or away from the operator.
- If the body is shifted towards the operator, smaller diameter of taper will be formed at tail stock end and if it is shifted away from the operator, bigger diameter of taper will be formed at tail stock end.
- Ball centres are used instead of ordinary lathe centres for proper bearing in the centre hole.
- Auto feed can be engaged.

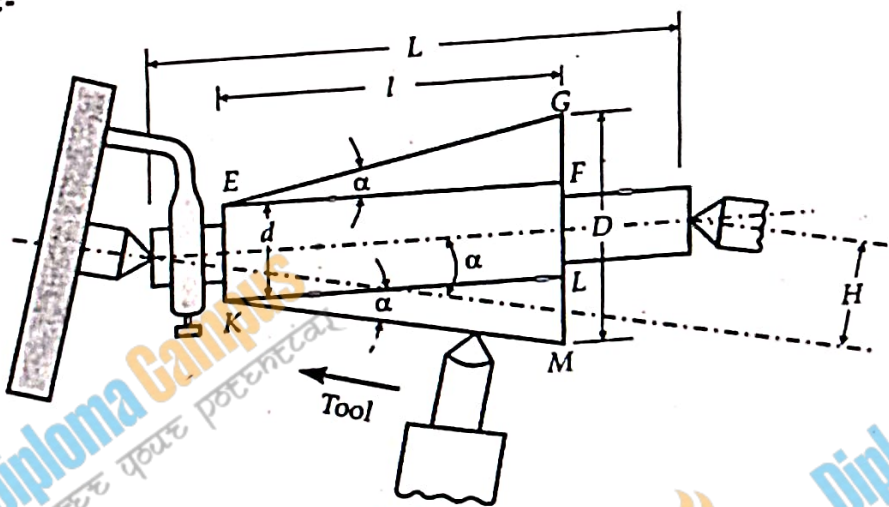
#### Advantages:-

1. Taper could be cut on maximum possible length.
2. External taper could be cut.
3. Auto feed could be engaged.
4. Thread on taper surface could be cut.
5. Finishing of taper is good.

#### Disadvantages:-

1. Steep taper could not be cut.
2. Internal taper could not be cut.
3. More time is consumed for setting and re-aligning of tail stock.
4. More skill is required.

#### Formulae :-



$$x = L \frac{D - d}{2l} = \text{Total length} \times \frac{\text{Total Taper}}{2 \times \text{Taper length}}$$

$x$  = the required setover in mm

$D$  = large dia. in mm

$d$  = small dia. in mm

$L$  = total length of work in mm

$l$  = length of tapered portion in mm.

In case the job is to be tapered over its full length,  $l$  will be  $= L$ . Therefore, the setover will be given by :

$$x = \frac{D - d}{2} = \frac{\text{Total Taper}}{2}$$

**Example 6.2.** Calculate the tail stock set-over for turning a taper on a job such that its two diameters are 80 mm and 50 mm. Total length of the job is 300 mm and the length of tapered portion is 200 mm only.

Now

$$\begin{aligned} x &= L \frac{D - d}{2l} \\ &= 300 \times \frac{80 - 50}{2 \times 200} \\ &= \frac{300 \times 30}{2 \times 200} \\ &= 22.5 \text{ mm.} \end{aligned}$$

**Example 6.3.** A drill sleeve has to carry an external taper of 1 in 20. Dia. of its large end is 50 mm and the length 80 mm. Find the amount of tail stock set-over.

$$\text{Taper on full length} = 80 \times \frac{1}{20} = 4 \text{ mm}$$

$$\therefore \text{Set-over } (x) = \frac{4}{2} = 2 \text{ mm.}$$

## 2.TAPER TURNING BY SWIVELLING COMPOUND REST

**Principle in setting:**— The job is set on lathe axis and the cutting tool travels at an angle.

**Amount of swivelling:** The compound rest is set at an angle equal to the half of the included angle of taper to be produced.

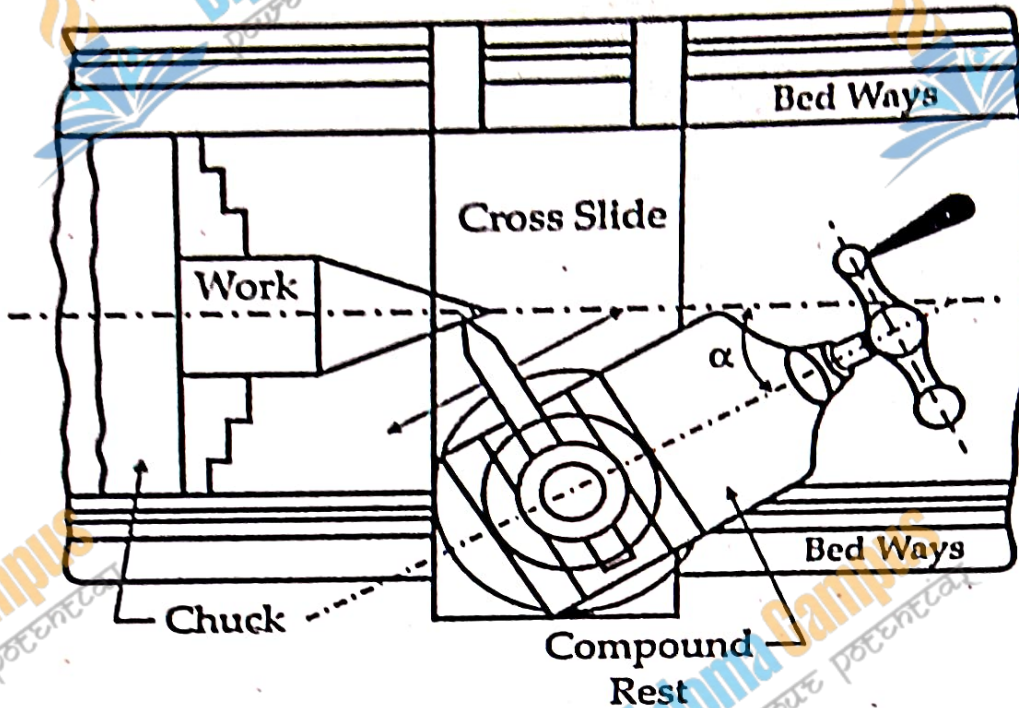
**Method:-**

- This method is applied for turning taper both externally and internally at any desired angle on a short length which is limited to the travel of top slide.
- At first the compound rest clamping nuts are loosen then it is set at required angle and the nuts are tightened.
- The cutting tool is set at centre height. While turning taper, top slide hand-wheel to be operated by hand.



Visual Watermark

- No automatic feed can be engaged.



#### Advantages:-

1. Taper can be produced at any desired angle.
2. Taper can be produced both externally and internally.
3. Setting time is very less.

#### Disadvantages :-

1. Taper length is restricted to the length of movement of top slide.
2. Auto feed could not be engaged.
3. Thread on taper could not be cut.
4. Finishing of taper surface is not so good due to hand feed.

#### Formulae :-

$$\tan \alpha = \frac{D - d}{2L}$$

Where, D=Big diameter  
d=Small diameter  
L=taper length  
α=Included angle

**Example 5.8** A Steel piece 100 mm long is to be taper turned for a length of 40 mm from its one end. Diameter of the work piece is 20 mm and the smaller end of the tapered part ends in a point. Find the angle through which the compound rest should be swivelled in order to turn the required taper.

**Solution.** Here  $D = 20$  ;  $d = 0$  ; and  $L = 40$

$$\tan \alpha = \frac{20}{2 \times 40} = 0.25$$

and  $\alpha = \tan^{-1} 0.25 = 14^{\circ} 3' \text{ (nearly).}$

**Example 5.9** Calculate the angle to which the compound rest should be swivelled in order to turn the taper on a job of which the finished sizes will be ; Larger dia. = 1" ; Smaller dia. 0.75" and Tapered length = 1.5".

**Solution.** Here

$$\tan \alpha = \frac{1.00 - 0.75}{2 \times 1.5}$$

$$= \frac{0.25}{3} = 0.0833$$

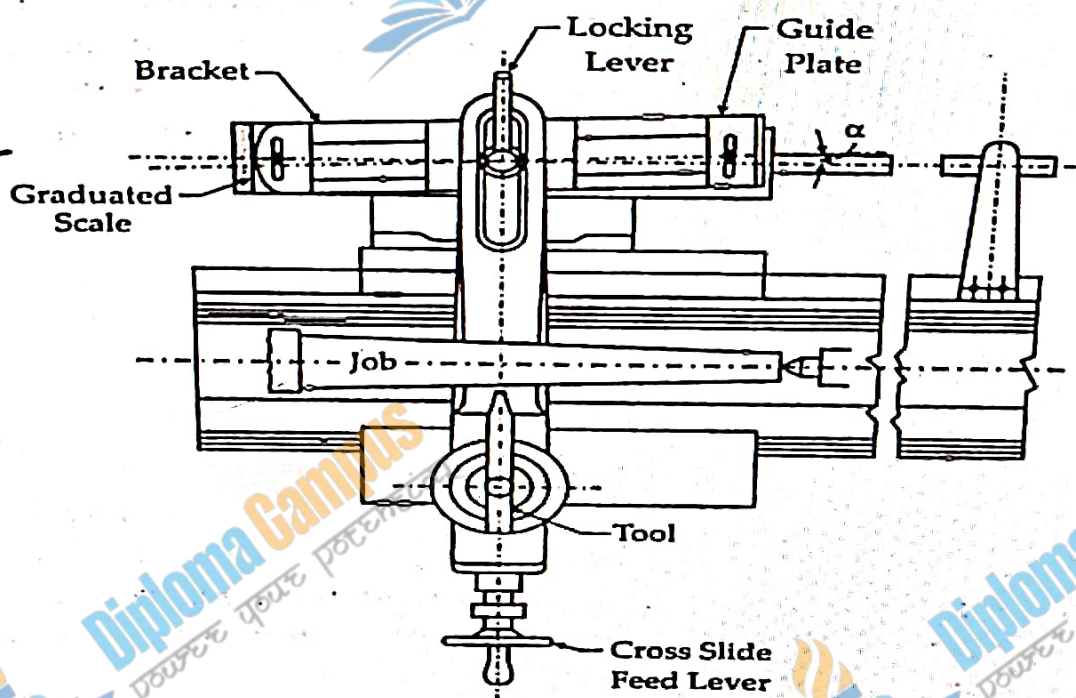
$$\alpha = \tan^{-1} 0.0833 = 4^{\circ} 46' \text{ (approx.)}$$

### 3. TAPER TURNING BY TAPER ATTACHMENT

**Principle in setting**

The job is set on a lathe-axis and the cutting tool moved at an angle.

**Construction of taper attachment:**



A taper turning attachment consists essentially of a bracket or frame which is attached at the rear end of the bed with the saddle and supports a guide bar pivoted at the centre. The bar is having graduations in degree at one end and taper per foot at other end. The bar may be swivelled on either side of the zero line and is set at desired angle with the lathe-axis and clamped by two nuts provided at each end of the bracket. There is a guide block or shoe which slides over the guide bar.

#### Method of setting :

1. Calculate the angle for swivelling the guide bar.
2. Swivel the guide bar by loosening the nuts and clamp it rigidly.
3. Lock the attachment with the bed at desired position.
4. Set the job on lathe axis either in a chuck or in between centres.
5. Turn taper either by hand feed or by engaging auto feed.

#### Advantages:-

1. Taper upto 24° included angle could be cut.
2. Taper could be cut both externally and internally.
3. Auto feed could be engaged.
4. Thread on taper surface could be cut.
5. Operations other than taper turning may be performed on a work piece in one setting without disturbing the attachment.

#### Disadvantages:-

1. Taper beyond 24 included angle could not be cut.
2. Taper length is restricted to guide bar length.
3. More time is required for setting the attachment if telescopic arrangement is not provided with the cross slide.

#### Formulae :-

$$\tan \alpha = \frac{D - d}{2L}$$

Where, D=Big diameter

d=Small diameter

L=taper length

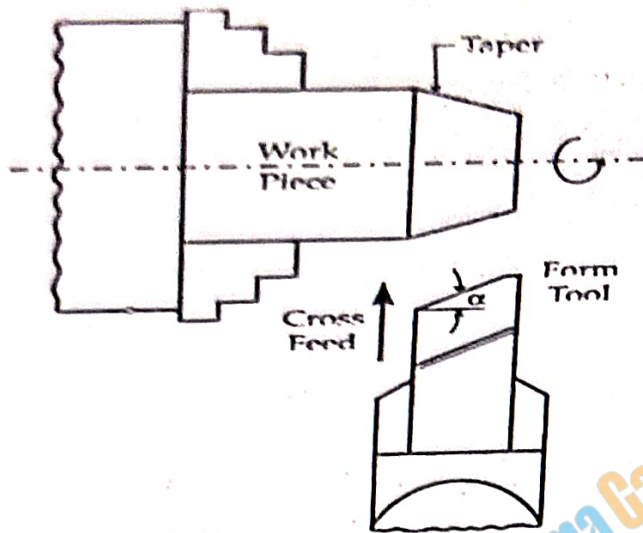
$\alpha$  =Included angle

#### 4.TAPER TURNING BY FORMED TOOL

##### Forming of cutting tool :-

A broad nose cutting tool is ground at an angle which is ground to the half of the included angle of taper to be produced.





#### setting of cutting tool:

The cutting tool is set at correct centre height and the axis of the cutting tool must be right angle to the axis of the job

#### Method:

The job will be revolving and the cutting tool is fed into the work.

#### Advantage-

1. Taper could be produced at any desired angle on short length.
2. Both external and internal taper could be produced.
3. Suitable for chamfering and bevelling on mass production.

#### Disadvantages :-

1. No precision taper could be produced.
2. Taper on larger length could not be cut.
3. Taper surface may have chattering marks.
4. Auto feed could not be engaged.

## 1. STEP AND TAPER TURNING

**AIM:** To perform facing, plain turning, step turning and taper turning on a given work piece of mild steel as per dimensions given in the figure.

### TOOLS REQUIRED:

- 1) Marking Block
- 2) Tool key
- 3) Chuck key
- 4) Single point cutting tool

### MEASURING INSTRUMENTS REQUIRED:

- 1) Steel rule
- 2) Out – side calipers
- 3) Vernier caliper

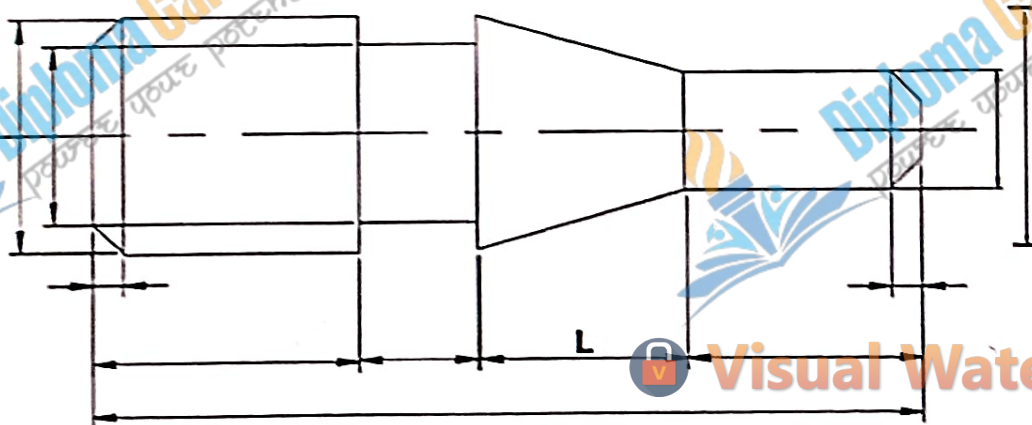
### MATERIAL REQUIRED:

A mild steel rod of length 100 x  $\phi$  25mm piece

### OPERATIONS:

- 1) Fixing the job on the chuck
- 2) Fixing the cutting tool in tool post
- 3) Facing
- 4) Plane turning
- 5) Step turning
- 6) Taper turning
- 7) Chamfering

### FIGURE:



## **PROCEDURE :**

1. With the help of marking block and chuck key, the job is fixed in the chuck so that the axis of job coincides with the axis of lathe.
2. The single point cutting tool is fixed on the tool post and sufficiently tightened with spanner.
3. Facing is done by giving cross feed with cross slide at one end.
4. Plain turning is done by giving longitudinal feed with carriage up to possible length of job.
5. The required taper angle is maintained by swiveling the compound slide.
6. The required chamfer is maintained.
7. The job is reversed & operations are repeated.
8. Required dimensions as per figure are maintained with accuracy using out side calipers, steel rule and Vernier calipers.
9. Chamfer is made as per figure.

## **TAPER ANGLE CALCULATIONS:**

$$\tan \alpha = \frac{D - d}{2L}$$

Where  $\alpha$  = Half taper angle

D = Major diameter of the work piece in mm

d = Minor diameter of the work piece in mm

L = Length of the taper in mm.

## **PRECAUTIONS :**

1. The job is tightly fixed in the chuck.
2. The cutting tool should be held rigidly on the tool post.

**RESULT :** Required job of given dimensions & shape is obtained.

